

# Notes: *Japan's Imperial Forest, Goryōrin, 1889–1946* and *The Lumber Industry in Early Modern Japan* (Conrad Totman)

Companion to [[totman\_green\_archipelago\_notes]]. Two more Totman books added to this folder:

- `japans-imperial-forest-goryorin-1889-1946_compress.pdf` (2007) — history of the Imperial Household's own forestry enterprise (Goryōrin), plus a stand-alone essay (Ch.8) on the 1871–76 "Kan/Min" (government/private) division of all Japanese woodland.
- `the-lumber-industry-in-early-modern-japan_compress.pdf` (1995) — the commercial/logistics side: how timber actually moved from stump to city, with a detailed case study of the Yamaguni (Kyoto-area) entrepreneurial lumber industry.

Read in full: Goryōrin Introduction, Ch.1–2 (background/rationale), Ch.5–7 (management plans, performance, financial payoff, assessment), Ch.8 + Epilogue (Kan/Min division). Lumber Industry Ch.2 (provisioning overview, skimmed), Ch.4 (Yamaguni case study, read closely), Ch.5 (Last Reflections). Ch.3 (transport mechanics) skimmed only.

## PART A — Goryōrin: a real large-scale sustained-yield forestry P&L (1889–1946)

This is the most concrete precedent in any of the three books for "professionally managed forest as a financial asset that took years to become profitable but then generated real, growing income." Directly useful if the CHP/DC project needs to show skeptical stakeholders that managed forestry has historically been bankable at scale in Japan, not just a subsidized cost center.

**What it was:** in 1889–90 the Meiji state carved out roughly 1.6 million chobu (~1.6M ha) of woodland as the personal income-producing estate of the Imperial Household, administered by its own Property Office (later the "Bureau of the Imperial Forest"). It ran as an independent forestry enterprise for 58 years until dissolved in 1947 when all Imperial property was nationalized.

### Financial trajectory — the headline number:

- Lost money in its first 1–2 years.
- Turned profitable from 1892.
- Annual net profit rose from ~234,000 yen (late 1890s) to ~6,000,000 yen (1920s–30s).
- By 1936–45 it generated **40% net profit margin** on gross income (vs. 23% for the much larger National Forest, kokuyurin) — better management and site selection paid off directly in margin, not just volume.
- Provided 19% of the Imperial Household's total income by 1916, 31% by 1940. Its cumulative asset value (stumpage + land + forest railroads) reached 2.6 billion yen by 1946 — 70% of the Household's total wealth.
- **Takeaway for citing to skeptical funders/village officials:** a real, historically-documented Japanese sustained-yield forestry operation went from loss to a 40%-margin, multi-decade profit center. This is a citable existence proof that "forestry as an income asset, not just a cost" is not a novel or foreign idea — it's Meiji-era Japanese state practice.

### Management structure (a template for what "well-run" forestry bureaucracy looked like):

- Central Property Office + regional headquarters (Sapporo/Hokkaido, Tokyo, Nagoya, Kiso) + branch offices + local section centers — a four-tier hierarchy matched to terrain.

- Staff grew from 33 (1889) to 1,829 by 1946 — i.e., real sustained-yield forestry at scale is a staffing-intensive enterprise, not a side project.
- Land was formally classified into three categories: **ringyogaichi** ("non-timber land" — peaks, cliffs, swamp, permanently excluded), **jun ringyochi** ("potential timberland" — undeveloped, targeted for improvement), **ringyochi** ("productive timberland" — the goal state). This three-tier classification is a clean, adoptable framework for triaging Sugano/village forest parcels by their realistic economic potential rather than treating all forest as equivalent.
- Rotation cycles were long: **60–120 years** from planting to final cut, organized into "rotation blocks" logged in sequence so the whole estate produced a continuous, staggered annual harvest once mature — the textbook definition of sustained yield.
- A concrete sustainable-harvest rule from the Sanru (Hokkaido) operations area: log ~4 parcels/decade, extracting **~33–50% of existing stumpage per cut**, calibrated so forest growth rate matched harvest rate. This is a usable rule-of-thumb ratio if the project needs to state a defensible sustainable off-take percentage for CHP fuel sourcing.

**Cautionary finding — the war years:** from 1943 onward, wartime demand forced the "Hundred Year Plan" to be formally suspended and the Property Office to authorize "special harvesting" — a deliberate reversion to exploitation-style logging (fullest possible cut from the most accessible sites) to meet military demand. Harvest jumped from 8.6M koku (1942) to 10.97M koku (1944), achieved mostly by *reducing* reforestation and aftercare spending. **This is the clearest documented case in either book of a demand spike overriding a sustained-yield plan and what that costs:** it left "a record of managerial planning and practice that had been reasonably successful" concealed under war-driven depredation, and post-war rebuilding kept cutting at unsustainable rates afterward. Directly relevant caution: any CHP fuel-supply contract needs a hard ceiling clause, because the historical pattern is that short-term demand pressure (whatever the cause) reliably breaks long-term forestry plans once it's allowed to.

**Why it eventually died (post-1947, and after):** after 1950, fossil fuels replaced firewood/charcoal, chemical fertilizer replaced green fertilizer, machinery replaced draft animals, and synthetic fiber replaced silk — the entire demand structure that made "people's woodland" (gen'ya) valuable to ordinary villagers evaporated within a couple of decades, and from the 1960s Japan's domestic lumber became uncompetitive against imports. The book's closing line: Japan's mountain sanrin plantations "mostly just stood there — and stand there today — aging... slowly losing their residual economic value... the woodland of Japan has ceased to be a resource worth fighting for." **This is the structural problem the CHP/AI-DC project is trying to reverse** — it is explicitly re-creating local demand for standing timber/biomass in exactly the kind of neglected mountain forest that lost its economic reason to exist after the 1950s energy transition.

## PART B — Kan/Min Division of Woodland, 1871–76 (Goryōrin Ch.8)

This is the single most useful section for **land-tenure/entity-structuring questions** the project keeps running into (forest cooperative access, Mitsue Kanko land, 丹羽製材 supply arrangements). Totman calls it "the greatest conflict between government and people in the entire history of Japan" — the Meiji state's attempt to force all of Japan's woodland (previously held under overlapping customary use-rights, iriaiken) into a clean binary: **kan'yuchi** (government land, tax-exempt, all shared-use rights extinguished) vs. **min'yuchi** (private land, taxable, owner-controlled).

**The core structural problem, in the government's own terms** (paraphrased from the book): rulers wanted **sanrin** — dense conifer high-forest for construction timber and charcoal — while villagers needed **gen'ya** — mixed scrubby woodland with understory, for fodder, fertilizer, food, and small construction wood. These are close to mutually exclusive land-use states (mature conifer canopy shades out the understory villagers depend on), so the tenure fight was really a fight over which of two incompatible ecological end-states a given hillside would become.

**This is structurally the same tension the Sugano reforestation plan has to navigate:** conifer/timber plantation vs. broadleaf coppice/wildlife-forage forest are not simultaneously optimizable on the same parcel — the project's own [[project\_reforestation\_principle]] (native broadleaf for wildlife forage) is implicitly choosing the "gen'ya" side of this 1870s argument, which is worth naming explicitly when talking to forestry-trained partners who may default to "sanrin" assumptions.

**The compromise category that actually solved the impasse — koyuchi:** rather than forcing every disputed parcel into pure government or pure private ownership, officials created a third bucket, **koyuchi** ("common-use"/"public" land) — for land that couldn't be cleanly surveyed, or that was known locally as village-shared woodland with real ongoing use-rights. Koyuchi wasn't a permanent legal category so much as an administrative "too complicated to force a clean split right now" holding pen — and it ended up covering more acreage than all of Japan's arable land combined. **Precedent worth citing:** Meiji reformers, when faced with irreconcilable dual claims on forest, didn't force a binary resolution — they created an explicit third, intermediate legal status. If Sugano/village forest tenure questions can't cleanly resolve into "NGO-owned" vs. "private landholder" vs. "village-owned," a koyuchi-style shared/communal designation with defined (not open-ended) use rights is a historically legitimate middle path, not an evasion.

**Final numeric outcome (by 1889):** government forest (kanrin + kan'yu rin'ya + goryorin combined) ended up at **53% of Japan Proper's woodland**, up from 31% in 1880 — i.e., even after a 15-year fight, the government only secured just over half; villagers/private holders retained the rest (min'yurin, ~8.4 million chobu). Even on the government's side, residual use-rights survived on many parcels, just heavily restricted and fee-bearing. **Nobody got a clean sweep** — worth remembering as a realistic expectation-setter if any Sugano tenure negotiation aims for a fully "clean" outcome; the historical base rate for these disputes is a messy, majority-but-not-total division with lingering secondary rights.

**A regulation table (Fig. 8-3) is a useful template** for how fast and how often policy actually changed hands during a live land-reform fight — six major reversals/adjustments in five years (1871–76). Useful context if the project needs to reassure partners that policy uncertainty during an active reform period is normal and doesn't indicate the process has failed.

**Officialdom's own diagnosed failure mode** (explicitly named in the text, worth citing verbatim in spirit): the reformers were mostly "urban samurai with little grasp of either village life or the existing woodland system," who "learned as they legislated, by trial and error," which "confused, frustrated, frightened, and angered people all across the realm." **This is a direct caution against designing Sugano land-use/tenure policy from Tokyo/urban/NGO-side assumptions without deep local consultation first** — it's the exact failure mode the Meiji forest reform is now studied as a cautionary example of.

## PART C — The Lumber Industry: entrepreneurial supply-chain economics (Yamaguni case study)

This book fills in the *commercial mechanics* that Green Archipelago and Goryōrin only sketch — genuinely useful if 丹羽製材/CHP fuel logistics need a template for how a real historical wood-supply chain organized itself without modern trucking or centralized ownership.

**The six-stage "hourglass" model of wood provisioning** (Ch.4, directly transferable as a supply-chain checklist):

1. Locating timber and arranging harvest rights
2. Felling and reducing to transportable form
3. Transport to a consolidation/storage point (the "neck" of the hourglass — the bottleneck stage)
4. Storage for curing and sale
5. Distribution to wholesaler → retailer/contractor/consumer
6. Disbursing income back through the chain

No single party controlled the whole chain — it required negotiated cooperation among multiple independent groups at each stage, financed by advances, and was "inherently risky" because commitments had to be made far ahead of any sale. This is a reasonable structural checklist for mapping out the Sugano-to-CHP fuel pathway (who controls felling rights → who processes/chips → who stores → who delivers to the plant → who gets paid when) and identifying where the "neck" bottleneck will actually sit.

**Guild/license structures as a control mechanism (kabu nakama, ikada han kabu):** Yamaguni's timber-producing households organized into a formally recognized guild (kabu nakama) holding tradeable, inheritable production/shipping rights (kabu). When newcomers (plantation growers, as opposed to old "axe-right" households) began threatening the old guild's control in the 1750s, the guild adapted by creating a **new, separate license class (ikada han kabu)** — rafting/shipping rights uncoupled from the ancestral axe-right — specifically so it could sell membership to successful new entrants and preserve collective price/market control. **This is a workable template if the forest cooperative / more trees / 丹羽製材 arrangement needs a mechanism to admit new landholders or wood suppliers into a coordinated pricing/logistics group without diluting or destroying the group's existing negotiating leverage.**

**Power shifted from urban merchants to rural producers over the Edo period, and why** — three advantages urban lumber merchants held in the 1600s (privileged government access, market know-how, buyer's-market pricing power from timber abundance) all eroded by the 1700s as timber grew scarce, government demand dropped, and rural literacy/organizational sophistication caught up. By the 1750s, rural lumbermen were "infiltrating and manipulating urban lumber markets to their own advantage," financing their own afforestation, and eventually becoming urban lumber magnates themselves. **Relevant reframe:** a CHP/DC project sourcing fuel from Sugano-area landholders is, historically, in the position of the *urban buyer* early in this cycle (holding the demand/capital advantage) — but the historical pattern is that this advantage is temporary and rural suppliers who organize (guilds, forward contracts, their own processing capacity) end up capturing more value over time. Worth designing supply contracts and 丹羽製材/village-cooperative relationships on the assumption that supplier leverage will grow, not stay fixed at today's terms.

**Nenkiyama confirmed again, with financing detail:** a Yamaguni timber merchant would buy a *standing young plantation* from its village-landholder grower, paying a down payment at purchase and the balance after logging — explicitly so the grower could recover part of his afforestation cost early while the merchant took on the risk of the growing period. This matches the nenkiyama mechanism already documented in [[totman\_green\_archipelago\_notes]] §3, with the added detail that it was one of several financing sources (alongside village mutual-aid societies (kō), moneylenders, and merchant cash advances/maegashi) — i.e., historically this was one tool in a mixed financing toolkit, not a standalone solution, which argues for combining a forward-supply contract with other financing legs (grants, cooperative capital, subsidy) rather than relying on it alone.

**Transport-tax/bottleneck realism:** even a well-organized, centuries-old timber river route (the Ōi/Katsura from Yamaguni to Kyoto, ~80km) suffered chronic problems worth naming as realistic risk factors for any Sugano-to-plant logistics plan: recurring silt/erosion damage requiring paid village labor to fix; a 5% transit tax at a fixed checkpoint; seasonal-only viable transport windows (spring/autumn, not midwinter); and a persistent bottleneck at the handoff point between rafting crews, that could leave timber sitting over a full season (with rot/value loss) if not cleared in time. **None of this is exotic** — it is the same category of problem (seasonal access, checkpoint/handoff friction, degradation while waiting) that a modern chip/pellet logistics chain from Sugano to a CHP plant will face, just with trucks and roads instead of rafts and rivers.

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## Summary table — new material beyond the first Green Archipelago note

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| Source                    | New concept                                                            | Modern application                                                                                           |
|---------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Goryōrin Ch.5–6           | 60–120yr rotation blocks, 33–50%/decade sustainable-harvest rule       | A defensible sustainable off-take ratio for CHP biomass sourcing                                             |
| Goryōrin Ch.6             | 40% net margin at scale by 1936–45                                     | Citable proof that managed Japanese forestry has historically been a real profit center, not just subsidized |
| Goryōrin Ch.6 (war years) | Demand-spike override of sustained-yield plan, 1943–47                 | Argues for a hard supply ceiling clause in any CHP fuel contract                                             |
| Goryōrin Ch.8             | sanrin vs. gen'ya as near-mutually-exclusive land states               | Names the tension already implicit in the broadleaf-forage reforestation choice                              |
| Goryōrin Ch.8             | koyuchi as a legitimate third tenure category                          | Template for a shared/communal designation if Sugano tenure can't cleanly resolve to owner/private           |
| Goryōrin Ch.8             | 53%/47% final kan/min split, messy and partial                         | Realistic expectation-setter for any tenure negotiation                                                      |
| Lumber Industry Ch.4      | Six-stage "hourglass" provisioning model                               | Checklist for mapping the Sugano→CHP fuel pathway and finding its bottleneck                                 |
| Lumber Industry Ch.4      | kabu nakama / ikada han kabu license restructuring                     | Template for admitting new suppliers into a coordinated cooperative without losing collective leverage       |
| Lumber Industry Ch.5      | Rural producers historically gain leverage over urban buyers over time | Design supply contracts assuming supplier leverage grows, not stays fixed                                    |

*Compiled from a direct read of both PDFs. Page-level citations available on request for grant or partner documents.*